

Matrix multiplication

Given $m \times p$ matrix A and $q \times n$ matrix B , as you recall, their product AB is defined if and only if $p = q$ and the resulting size of the matrix product AB is $m \times n$.

Definition 5. Let $A = (a_{ij})$ and $B = (b_{ij})$ be $m \times p$ and $p \times n$ matrices, respectively. We define their product AB as follows - if we use c_{ij} for $1 \leq i \leq m$ and $1 \leq j \leq n$ to denote the entries of the product AB , that is, $AB = (c_{ij})$, we have $c_{ij} = \sum_{k=1}^p a_{ik}b_{kj}$.

For each $1 \leq i \leq m$ and $1 \leq j \leq n$, the entry c_{ij} in the product is the dot product of the i -th row vector in the first matrix and j -th column vector in the second matrix.

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ip} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mp} \end{pmatrix} \begin{pmatrix} b_{11} & \cdots & b_{1j} & \cdots & b_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ b_{p1} & \cdots & b_{pj} & \cdots & b_{pn} \end{pmatrix} = \begin{pmatrix} c_{11} & \cdots & c_{1j} & \cdots & c_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ c_{i1} & \cdots & c_{ij} & \cdots & c_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ c_{m1} & \cdots & c_{mj} & \cdots & c_{mn} \end{pmatrix}$$

Although we shall give a detailed example, we expect you to be fluent with multiplying matrices from your first year mathematics.

Example Below we are looking at the product of a 2×3 matrix and a 3×2 matrix and so the result will be a 2×2 .

$$\begin{pmatrix} 0 & 1 & 4 \\ 2 & 7 & 6 \end{pmatrix} \begin{pmatrix} 0 & 10 \\ 3 & 0 \\ 5 & 7 \end{pmatrix}$$

Let $\begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix}$ denote the resulting matrix. Its entries will then be give by

$$c_{11} = 0 \cdot 0 + 1 \cdot 3 + 4 \cdot 5 = 23$$

$$c_{12} = 0 \cdot 10 + 1 \cdot 0 + 4 \cdot 7 = 28$$

$$c_{21} = 2 \cdot 0 + 7 \cdot 3 + 6 \cdot 5 = 51$$

$$c_{22} = 2 \cdot 10 + 7 \cdot 0 + 6 \cdot 7 = 62$$

That is,

$$\begin{pmatrix} 0 & 1 & 4 \\ 2 & 7 & 6 \end{pmatrix} \begin{pmatrix} 0 & 10 \\ 3 & 0 \\ 5 & 7 \end{pmatrix} = \begin{pmatrix} 23 & 28 \\ 51 & 62 \end{pmatrix}$$

We now show that the matrix multiplication is associative.

Side note: Later we shall think of matrices as representing some of functions, and that matrix multiplication corresponds to composition of those functions.

Indeed, function composition is associative:

Let X, Y, Z, W be sets, given functions $f : X \rightarrow Y$, $g : Y \rightarrow Z$ and $h : Z \rightarrow W$, we consider the two functions $h \circ (g \circ f) : X \rightarrow W$ and $(h \circ g) \circ f : X \rightarrow W$ and show that they are equal. To show that they are equal, we show that they are equal on each input, that is, for each $x \in X$, $(h \circ (g \circ f))(x) = h((g \circ f)(x)) = h(g(f(x))) = (h \circ g)(f(x)) = ((h \circ g) \circ f)(x)$, these equalities hold for each $x \in X$ and so $h \circ (g \circ f) = (h \circ g) \circ f$.

However, below we shall show matrix multiplication is associative using its definition.

Proposition 3. Let $A = (a_{ij})$, $B = (b_{ij})$ and $C = (c_{ij})$ be $m \times p$, $p \times q$ and $q \times n$ matrices, respectively. Then $A(BC) = (AB)C$, that is the matrix multiplication is associate.

Proof. We first notice that both $A(BC)$ and $(AB)C$ have the same size, that is $m \times n$, we are left to show that their corresponding entries are equal. We begin with the entry on the i -th row and j -th column of matrix $A(BC)$ and show that it is equal to the entry on the i -th row and j -th column of matrix $(AB)C$, that is:

$$\begin{aligned} \sum_{k=1}^p a_{ik} \left(\sum_{l=1}^q b_{kl} c_{lj} \right) &= \sum_{k=1}^p \sum_{l=1}^q a_{ik} (b_{kl} c_{lj}) \\ &= \sum_{k=1}^p \sum_{l=1}^q (a_{ik} b_{kl}) c_{lj} \\ &= \sum_{l=1}^q \sum_{k=1}^p (a_{ik} b_{kl}) c_{lj} \\ &= \sum_{l=1}^q \left(\sum_{k=1}^p (a_{ik} b_{kl}) \right) c_{lj} \end{aligned}$$

and so $A(BC) = (AB)C$. □

What about commutativity? Is it true that for all $m \times n$ matrix A and $n \times m$ matrix B we have $AB = BA$?

The answer is no! indeed, if A is a 3×2 and B a 2×3 matrix, then AB has size 3×3 while BA has size 2×2 and so the two cannot be equal.

Even in the case when their sizes are equal, we still have a counter-example.

Consider:

$$\begin{pmatrix} 0 & 1 \\ 2 & 7 \end{pmatrix} \begin{pmatrix} 0 & 10 \\ 3 & 0 \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 21 & 20 \end{pmatrix} \text{ while } \begin{pmatrix} 0 & 10 \\ 3 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 2 & 7 \end{pmatrix} = \begin{pmatrix} 20 & 70 \\ 0 & 3 \end{pmatrix}$$

Matrix multiplication and addition do satisfy the left and right distribute laws.

Proposition 4. Let $A = (a_{ij})$ be an $m \times p$ matrix, $B = (b_{ij})$ and $C = (c_{ij})$ $p \times q$ matrices and $D = (d_{ij})$ a $q \times n$ matrix. Then we have

- $A(B + C) = AB + AC$; (left distributivity)
- $(B + C)D = BD + CD$. (right distributivity)

Proof. We shall only prove one, the left distributivity, the other one can be proved similarly and we leave it as an exercise. Note that we cannot use commutativity to deduce one from the other.

Now, we notice that $A(B + C)$ and $AB + AC$ both have the same size, that is, $m \times q$. We now show that their corresponding entries are equal. We begin with the entry on the i -th row and j -th column of matrix $A(B + C)$ and show that it is equal to the entry on the i -th row and j -th column of matrix $AB + AC$, that is:

$$\begin{aligned}\sum_{k=1}^p a_{ik}(b_{kj} + c_{kj}) &= \sum_{k=1}^p (a_{ik}b_{kj} + a_{ik}c_{kj}) \\ &= \sum_{k=1}^p a_{ik}b_{kj} + \sum_{k=1}^p a_{ik}c_{kj}\end{aligned}$$

and so $A(B + C) = AB + AC$. □

Exercises:

1. Simplify $\begin{pmatrix} 0 & 7 \\ 3 & 0 \end{pmatrix} \left[\begin{pmatrix} 0 & 5 \\ 3 & 1 \end{pmatrix} + 4 \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix} \right]$
2. Are there real numbers λ and μ such that $\lambda \begin{pmatrix} 0 & 1 \\ 4 & 1 \end{pmatrix} + \mu \begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$?